

# A Short Proof of an Equivalence Statement

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## 1 Introduction

In this document, I provide a concise equivalence proof on the topic of filters and ultrafilters from the field of model theory. The theorem was part of my course and I completed the proof from class for the sake of presenting a sample of a mathematical proof.

The presentation follows the typical logical flow used in research papers: formal definition, theorem statement, and a two-part proof (*if* and *only if*).

## 2 Preliminaries

**Definition.** Let  $I$  be a non-empty set. A filter  $\mathcal{F}$  on  $I$  is a collection of subsets of  $I$  with the following properties:

1.  $\emptyset \notin \mathcal{F}$
2.  $I \in \mathcal{F}$
3. for all  $A, B \subseteq I$ , if  $A, B \in \mathcal{F}$  then  $A \cap B \in \mathcal{F}$
4. for all  $A, B \subseteq I$ , if  $A \in \mathcal{F}$  and  $A \subseteq B$  then  $B \in \mathcal{F}$

**Definition.** An ultrafilter  $\mathcal{U}$  is a maximal filter, meaning it can not be properly extended while remaining a filter.

## 3 Main Result

**Theorem.** If  $\mathcal{F}$  is a filter, the following statements are equivalent:

1.  $\mathcal{F}$  is an ultrafilter;
2. for all  $A \subseteq I$ , either  $A \in \mathcal{F}$  or  $A^C \in \mathcal{F}$ .

*Proof.* We will prove both directions of the equivalence separately.

First we will show that **1**  $\implies$  **2**. Assume  $\mathcal{F}$  is an ultrafilter on  $I$ . We want to show that for all  $A \subseteq I$ , either  $A \in \mathcal{F}$  or  $A^C \in \mathcal{F}$ .

Assume  $A \subseteq I$  and  $A \notin \mathcal{F}$ . We will show that  $A^C \in \mathcal{F}$ .

Since  $A \notin \mathcal{F}$ , we can define a filter  $\mathcal{G}$  that extends  $\mathcal{F}$  by  $A$ :

$$\mathcal{G} := \{X \subseteq I \mid \exists B_1, \dots, B_n \in \mathcal{F} \cup \{A\} \text{ s.t. } B_1 \cap \dots \cap B_n \subseteq X\}.$$

Since  $\mathcal{F}$  is an ultrafilter, and  $\mathcal{G}$  is a proper extension of  $\mathcal{F}$ , it follows that  $\mathcal{G}$  cannot be a filter.

But we can observe that  $\mathcal{G}$  fulfills properties 2, 3, and 4 from our definition of a filter:

2.  $I \in \mathcal{G}$  because  $I \in \mathcal{F}$  and  $\mathcal{F} \subset \mathcal{G}$ .
3. If  $A', B' \in \mathcal{G}$  then  $A' \cap B' \in \mathcal{G}$ . This follows directly from our definition of  $\mathcal{G}$  being closed under intersection.
4. If  $A' \in \mathcal{G}$  and  $A' \subseteq B'$  then  $B' \in \mathcal{G}$ . This also follows directly from our definition of  $\mathcal{G}$  having upwards closure.

Since  $\mathcal{G}$  is not a filter, but still fulfills properties 2, 3, and 4 from our definition, we conclude that property 1 must fail.

Therefore  $\emptyset \in \mathcal{G}$  and it must then be the case that

$$A \cap (B_1 \cap \dots \cap B_n) = \emptyset \text{ for some } B_i \in \mathcal{F}.$$

Note that we can ignore occurrences of  $A$  in our list of  $B$ s since  $A \cap A = A$ .

Since all the  $B_i$  are in  $\mathcal{F}$ , by the definition of a filter,  $C := \bigcap_{i=1}^n B_n \in \mathcal{F}$ .

Since  $A \cap C = \emptyset$ , it follows that  $C \subseteq A^C$ . And since  $C \in \mathcal{F}$ , it follows from the definition of a filter that  $A^C \in \mathcal{F}$ .

Therefore for all  $A \subseteq I$ , either  $A \in \mathcal{F}$  or  $A^C \in \mathcal{F}$ .

Now we will show that **2**  $\implies$  **1**. Assume that for all  $A \subseteq I$ , either  $A \in \mathcal{F}$  or  $A^C \in \mathcal{F}$ . We want to show that  $\mathcal{F}$  is a maximal filter.

Let  $\mathcal{G}$  be a filter such that  $\mathcal{F} \subseteq \mathcal{G}$ . It suffices to show that  $\mathcal{F} = \mathcal{G}$ . To this end, suppose that  $A \in \mathcal{G}$  and we will show that necessarily  $A \in \mathcal{F}$ . Since  $\mathcal{G}$  is a filter,  $A \cap A^C = \emptyset \notin \mathcal{G}$ . As  $A \in \mathcal{G}$ , it follows that  $A^C \notin \mathcal{G}$ . Therefore also  $A^C \notin \mathcal{F}$ . By our assumption, it follows that  $A \in \mathcal{F}$ .

Since any  $A \in \mathcal{G}$  is also in  $\mathcal{F}$ , we have  $\mathcal{F} = \mathcal{G}$  and  $\mathcal{F}$  is indeed an ultrafilter, as there are no proper extensions of  $\mathcal{F}$ .

This completes the proof. □

## 4 Conclusion

I have provided the reader with a proof of an equivalence in the field of model theory, specifically on the topic of filters. This short proof should be representative of the types of proofs I engage with during my academic studies.